

ASHIYA AB, Japan

WMO: 478030

Lat: 33.883N

Lon: 130.653E

Elev: 30

StdP: 100.97

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	0.2	-8.1	1.9	2.9	-7.0	2.1	3.6	12.3	3.2	11.2	4.4	4.4	160	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	32.8	25.8	31.2	25.8	30.2	25.5	26.7	30.2	26.2	29.7	25.9	29.3	5.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.4	29.0	25.2	20.4	28.3	25.0	20.1	28.1	84.2	30.6	82.3	30.0	80.4	29.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.1	8.8	7.9	-2.5	34.4	0.8	0.9	-3.1	35.1	-3.6	35.6	-4.1	36.1	-4.7	36.7		
			-3.8	27.7	0.9	0.6	-4.4	28.2	-5.0	28.6	-5.5	28.9	-6.1	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	5.7	6.4	9.2	13.8	18.3	22.0	26.2	27.4	23.5	18.2	12.8	7.9
	DBStd	7.87	2.39	2.86	3.00	3.00	2.43	2.21	2.26	1.89	2.48	2.64	3.05	2.77
	HDD10.0	383	136	107	51	4	0	0	0	0	0	0	8	76
	HDD18.3	1703	393	334	282	139	31	1	0	0	0	34	166	323
	CDD10.0	2577	2	6	27	118	257	360	502	539	405	256	93	12
	CDD18.3	854	0	0	0	3	29	111	243	281	155	31	1	0
	CDH23.3	6709	0	0	1	16	112	449	2190	2902	956	83	2	0
	CDH26.7	1979	0	0	0	2	11	74	682	1026	181	4	0	0
Wind	WSAvg	3.7	4.3	4.2	4.1	3.7	3.4	3.2	3.1	3.4	3.5	3.6	3.7	4.3
Precipitation	PrecAvg	1775	76	84	111	133	152	283	317	187	177	85	95	74
	PrecMax	2412	219	152	267	237	319	508	684	435	501	227	174	135
	PrecMin	1041	30	34	58	36	35	41	61	25	68	16	28	20
	PrecStd	323	39	30	46	56	74	130	186	108	95	55	44	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	18.8	20.9	25.8	28.0	30.8	33.2	34.2	31.9	27.1	22.8	17.1
		MCWB	12.2	14.6	14.0	18.4	19.2	24.9	25.8	25.9	25.3	21.7	17.6	14.0
	2%	DB	12.8	14.9	18.1	22.2	26.0	28.8	32.0	33.1	30.1	25.2	20.8	15.2
		MCWB	9.8	11.9	12.7	15.8	18.5	23.2	25.8	25.9	25.1	20.4	16.7	11.6
	5%	DB	11.2	13.0	16.1	20.8	24.2	27.1	31.0	32.1	28.8	24.1	19.1	14.1
		MCWB	8.0	9.7	12.1	15.3	17.9	22.3	25.8	25.7	24.3	19.6	15.3	10.6
	10%	DB	10.1	11.2	14.2	19.0	23.0	26.0	30.1	31.0	27.2	22.9	18.1	13.0
		MCWB	7.0	8.1	10.9	14.4	17.5	22.0	25.6	25.8	23.4	18.7	14.6	9.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	15.0	16.0	20.1	22.7	25.6	27.1	27.6	26.4	23.5	19.2	15.2
		MCDB	14.5	17.8	18.0	23.4	25.0	29.1	30.7	30.9	29.5	25.8	20.5	16.3
	2%	WB	10.6	12.7	14.1	17.7	20.8	24.7	26.5	27.0	25.7	22.0	17.5	12.7
		MCDB	11.8	14.4	16.7	20.7	23.2	27.6	30.0	30.5	28.7	24.4	19.7	14.5
	5%	WB	8.7	10.6	12.8	16.4	19.6	23.8	26.1	26.5	25.0	20.7	16.4	11.4
		MCDB	10.7	12.2	15.5	19.3	22.4	26.0	29.6	30.0	27.7	22.9	18.6	13.4
	10%	WB	7.3	8.9	11.5	15.3	18.6	22.9	25.8	26.1	24.3	19.5	15.2	10.2
		MCDB	9.4	11.1	14.1	18.0	21.7	24.9	29.1	29.6	26.8	22.1	17.7	12.3
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.1	7.7	8.1	7.7	6.1	5.8	6.3	6.3	7.6	7.7	7.1
		MCDBR	8.7	9.8	10.5	10.9	10.3	8.3	7.3	7.9	7.4	8.6	8.9	8.8
		MCWBR	6.4	7.3	7.0	6.2	5.2	4.1	3.1	2.9	3.4	4.7	5.7	6.3
	5% WB	MCDBR	7.7	8.3	8.8	8.2	7.5	6.1	6.5	6.8	6.2	6.3	7.2	7.4
MCWBR		6.7	7.2	6.8	6.8	5.5	5.1	3.8	3.1	3.0	3.3	4.2	5.4	6.3
Clear-Sky Solar Irradiance	taub	0.430	0.471	0.533	0.554	0.560	0.575	0.570	0.533	0.503	0.458	0.431	0.422	
	taud	2.058	1.985	1.838	1.842	1.889	1.937	1.997	2.100	2.134	2.156	2.142	2.093	
	Ebn at Noon	745	758	744	753	753	740	741	761	765	765	741	726	
	Edh at Noon	132	158	198	206	199	189	177	158	146	132	121	121	
All-Sky Solar Radiation	RadAvg	2.15	2.88	3.97	4.82	5.27	4.54	4.90	5.14	4.09	3.46	2.56	2.01	
	RadStd	0.20	0.29	0.28	0.40	0.60	0.50	0.75	0.57	0.41	0.36	0.24	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page